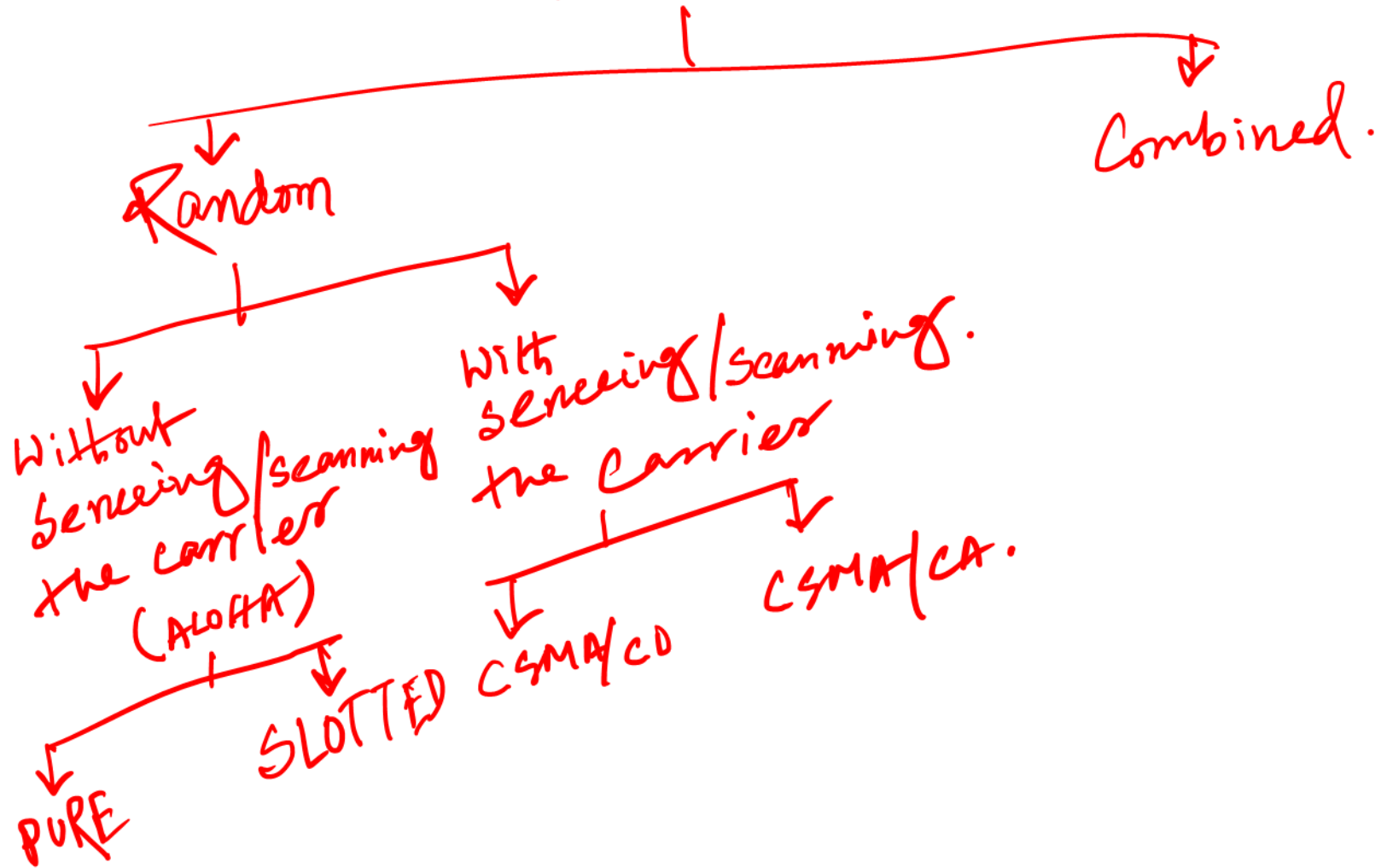
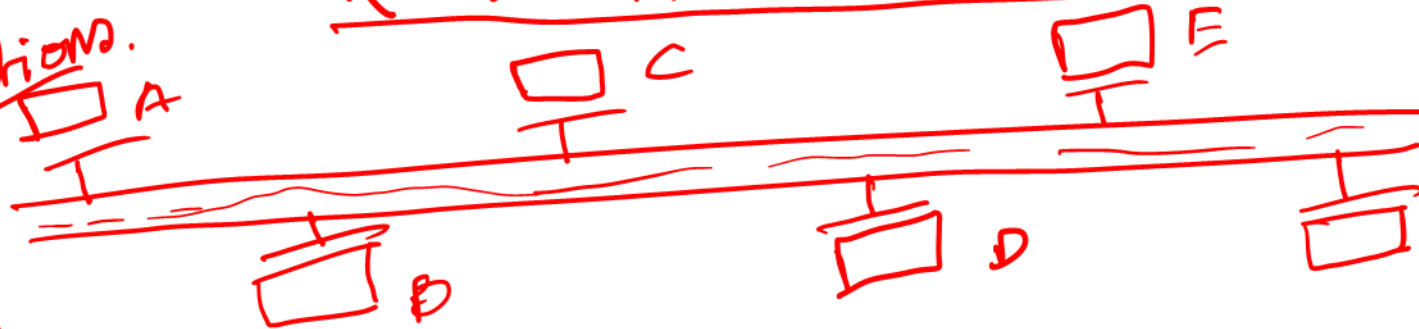


MAC (Media Access Control)



Random Access Protocol.

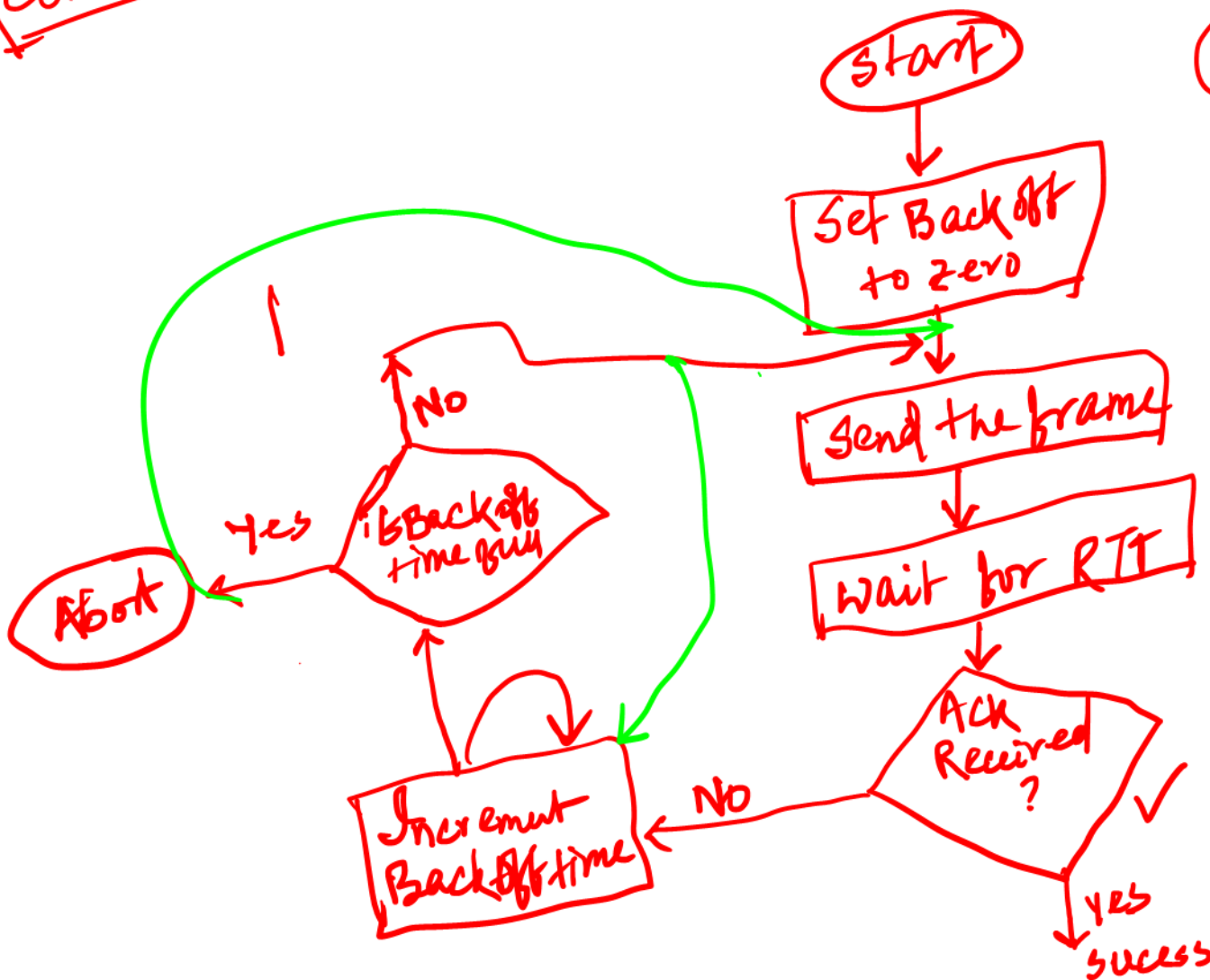
stations.



data transmit
at any
time.

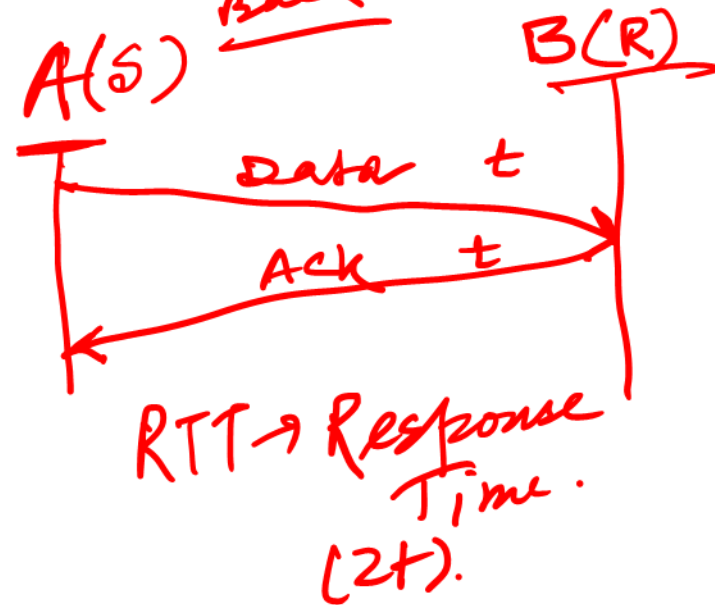
Collision

Address



$$(2t) + (5s)$$

Back off.



✓ ALOHA (RAP):

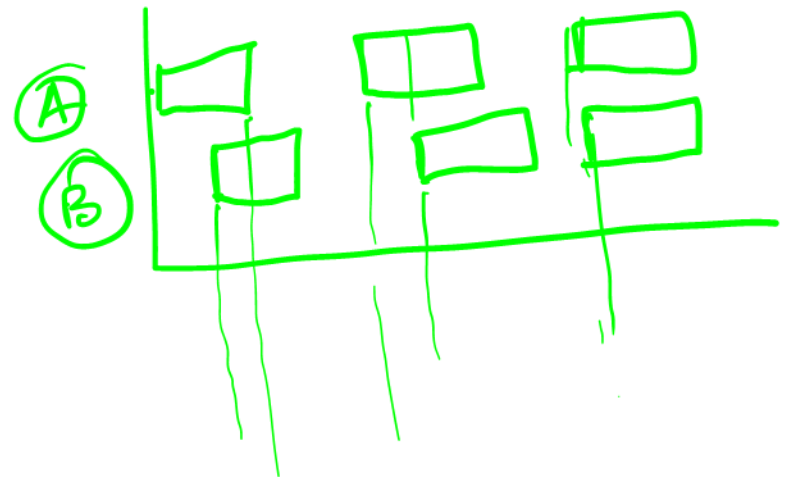
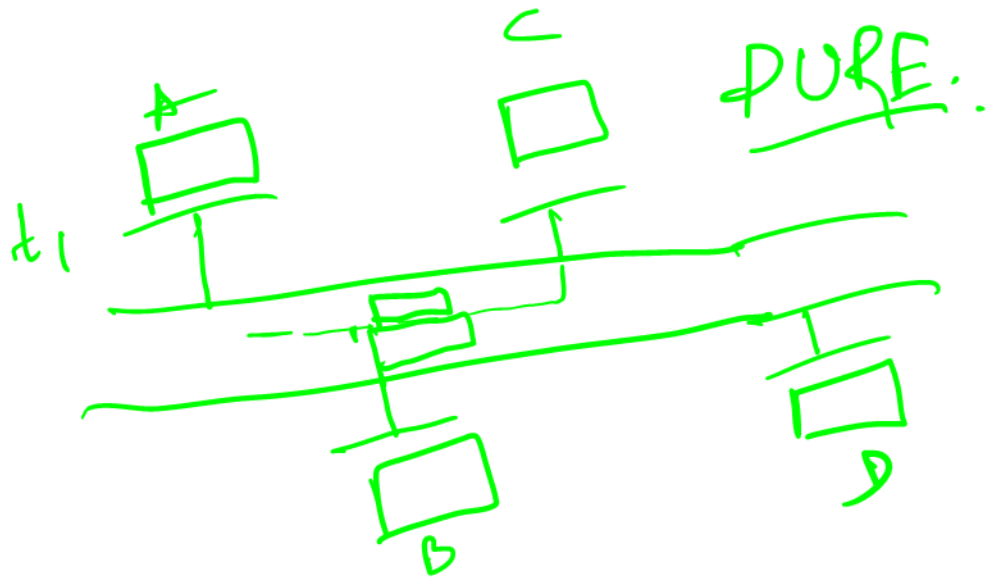
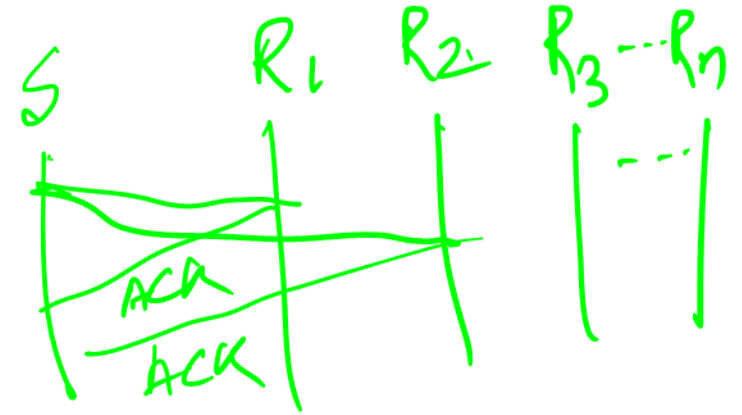
(i) Multiple Access.

(ii) ACK - //

ALOHA

Pure.

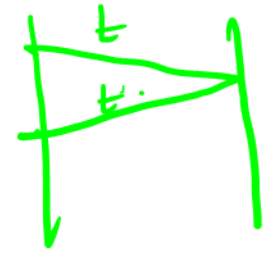
Slotted.



Variable Time $\Rightarrow$



$$(2t + t_0)$$



$$2t$$

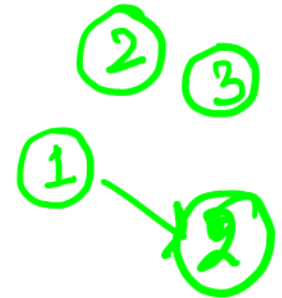


Efficiency (η) $\Rightarrow$ $G \cdot e^{-2G}$

$$\frac{dt}{dG} \Rightarrow 0.184$$

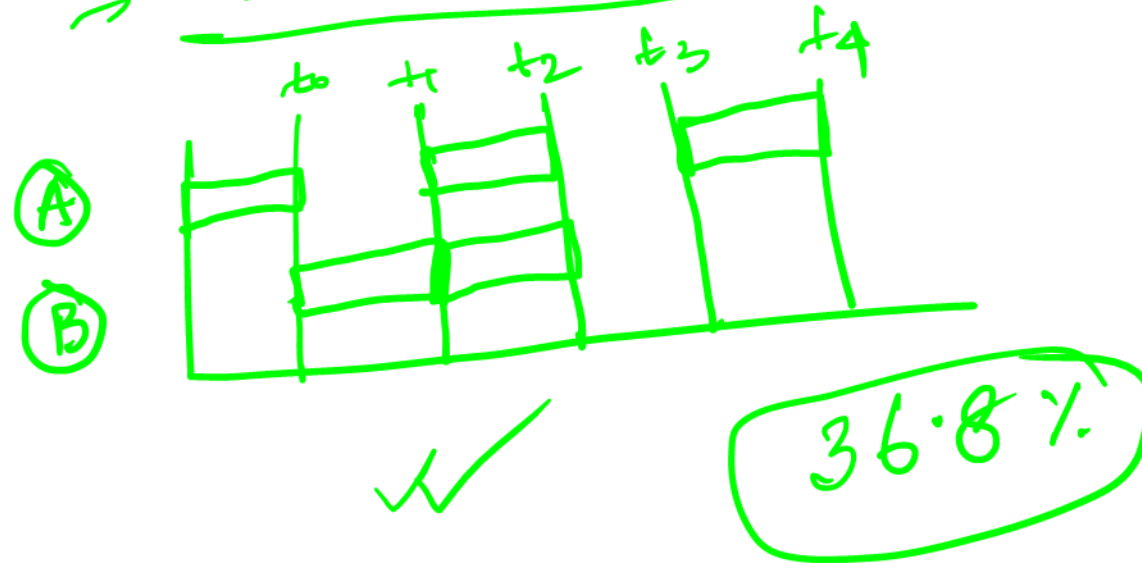
$$18.4\%$$

$$\frac{1}{2}$$



2

→ SLOTTED ALOHA

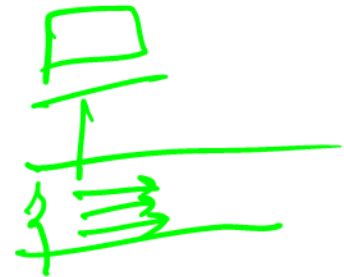
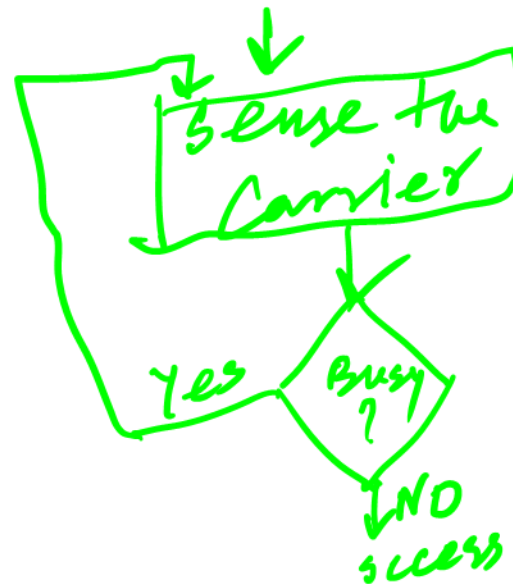


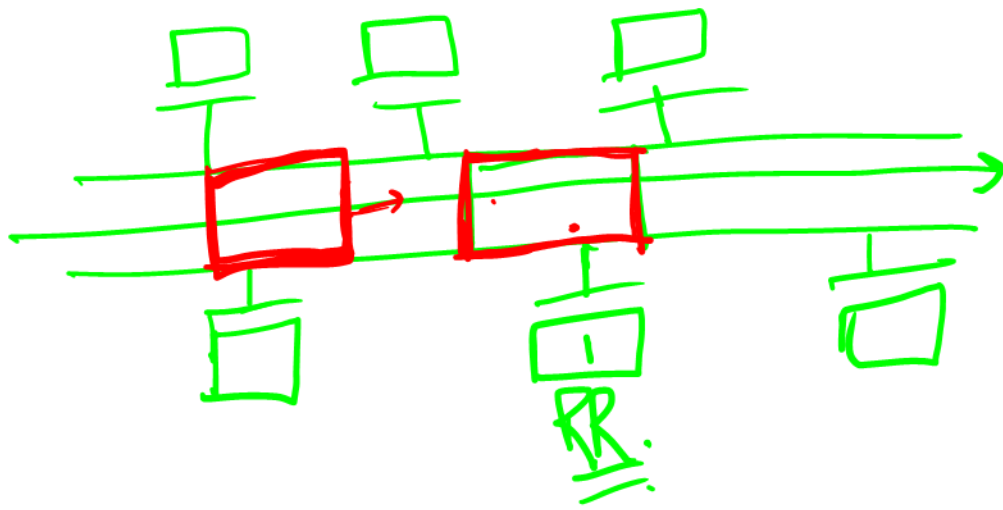
CSMA-CD (Carrier Sense Multiple Access - Collision Detection)

Persistence

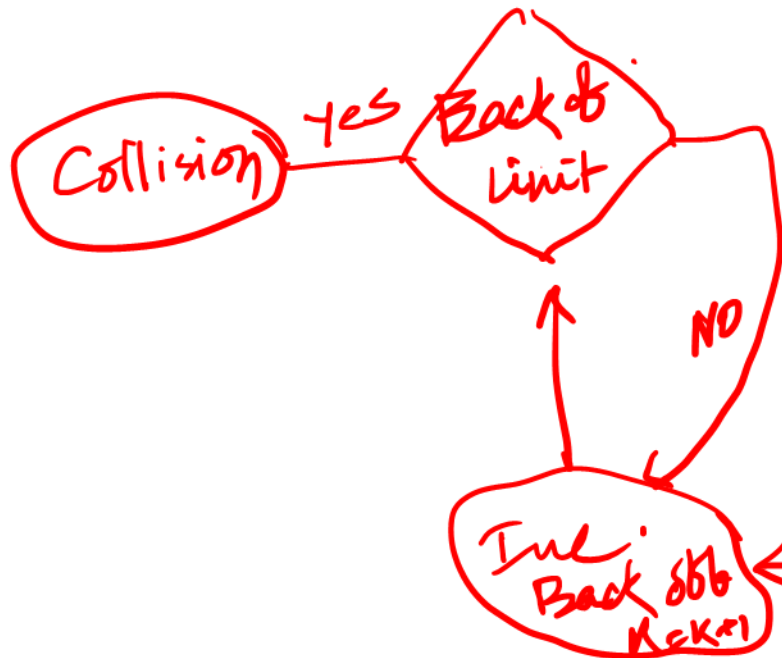
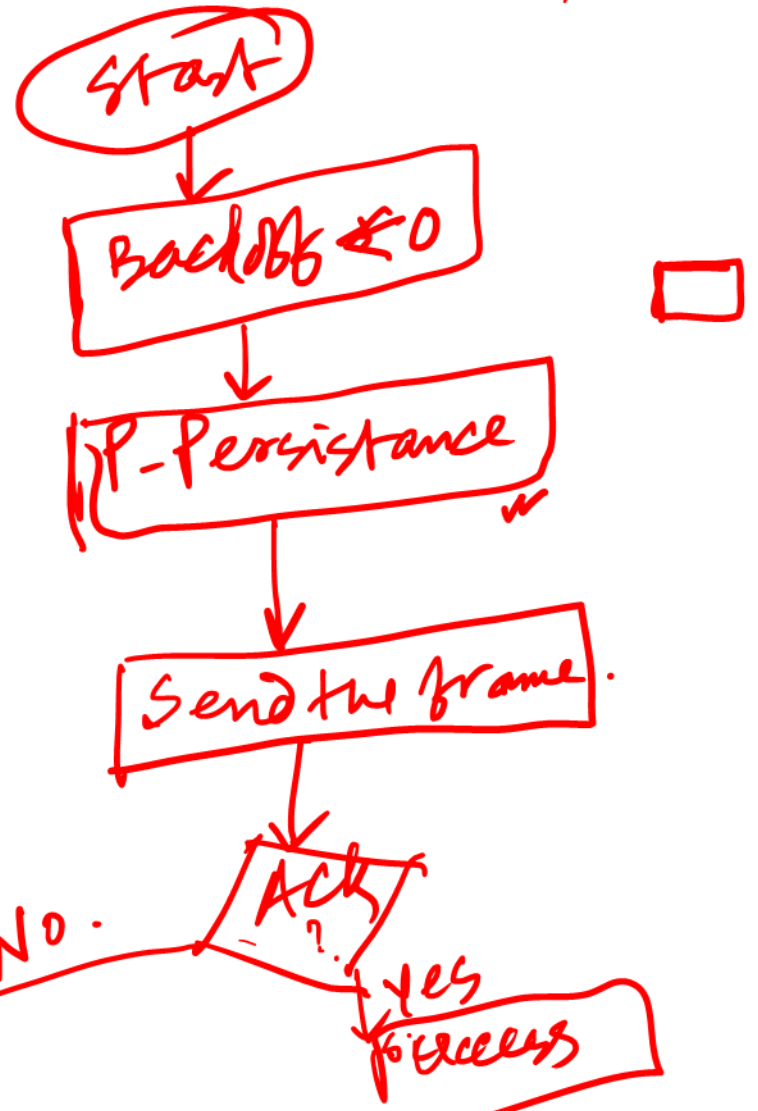
→ $\downarrow$ Persistence
→ P-Persistence.

Token Ring
Token Bus





P-Persistence
 ↓
 Collision with
 Probability - P.



CSMA-CA

Collision Avoidance

